

Project Report

**“Youth Distress Analytics: Statistical
and Machine Learning Study of
Maturity, Confusion, and Gender”**

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Abstract

Youth in their 20s often face high levels of uncertainty due to exams, placements, and relationship decisions, which can increase psychological distress. This project analyzes the Young People Survey dataset (Kaggle), focusing on 564 respondents aged 20–29, to study how psychological maturity and confusion relate to distress, and whether gender and internet usage play a role.

Using Python, I constructed composite scores for maturity, confusion, and distress from multiple questionnaire items on responsibility, decision-making, mood swings, loneliness, happiness, energy, and health. I then performed exploratory data analysis, hypothesis testing (regression, t-tests, ANOVA), clustering, and basic machine learning.

The results show that confusion is a strong positive predictor of distress, explaining around 20% of its variance in a simple regression, while maturity has only a weak effect once confusion and gender are controlled. Female respondents report significantly higher distress than males, but simple categories of internet usage do not show significant differences. Cluster analysis identifies three profiles, including a high-maturity but high-confusion and high-distress group, highlighting confusion as a key risk factor. This project demonstrates end-to-end survey data analysis, statistical modeling, and machine learning skills relevant for analytics roles.

3. Introduction

3.1. Background

The twenties are often described as a “time of confusion” because young adults face multiple simultaneous pressures: competitive exams, campus placements, financial independence, and relationship decisions. These pressures can trigger confusion about life direction and increase psychological distress. Understanding how maturity and confusion interact with distress in this age group is important both for mental health research and for designing support systems in universities.

3.2. Objective

The aim of this project is to quantitatively analyze the relationships between psychological maturity, confusion, and distress among youth aged 20–29 using real survey data, and to identify distinct psychological profiles using clustering and machine learning.

1. Construct composite indices for maturity, confusion, and distress from survey items.
2. Test hypotheses about the effects of maturity, confusion, gender, and internet usage on distress.
3. Identify clusters of youth based on maturity, confusion, and distress scores.
4. Demonstrate end-to-end analysis using Python .

3.3 Research Questions

1. Is higher maturity associated with lower distress among youth in their 20s?
2. Is higher confusion associated with higher distress?
3. Are there significant gender differences in distress?
4. What psychological profiles emerge from clustering based on maturity, confusion, and distress?

4. Data Description and Variable Construction

4.1 Dataset overview

The project uses the Young People Survey dataset available on Kaggle. The original survey contains responses from youth aged 15–30 on multiple topics such as music preferences, movies, hobbies, attitudes, fears, health, and demographics. For this study, I filtered the dataset to respondents aged 20–29, resulting in a final sample of 564 observations.

4.2 Selected variables

From the full questionnaire, I selected items related to responsibility, planning, decision-making, mood swings, loneliness, happiness, energy, health, gender, age, and internet usage. These items were grouped into three conceptual indices: maturity, confusion, and distress.

Category	Items
Maturity	Prioritising workload, Thinking ahead, Final judgement, Reliability, Keeping promises, Achievements, Responding to a serious letter, Assertiveness, Punctuality_num
Confusion	Decision making, Judgment calls, Mood swings, New environment, Life struggles
Distress	Loss of interest, Loneliness, Life struggles, Happiness in life, Energy levels, Health

4.3 Encoding and scoring

Some questionnaire responses were stored as text categories (e.g., “i am always on time”, “i am often running late”) or descriptive phrases for internet usage. These were converted into numeric codes to allow statistical modeling. For example, punctuality was mapped to a three-level numeric scale from low to high punctuality, and internet usage was coded from 0 (“no time at all”) to 3 (“most of the day”). Gender was encoded as 0 for male and 1 for female.

Composite indices were computed as the average of conceptually related items:

Maturity_score: average of responsibility, planning, reliability, punctuality, assertiveness, and related items.

Confusion_score: average of decision difficulty, judgment calls, mood swings, new environment, life struggles.

Distress_score: average of loss of interest, loneliness, lower happiness, lower energy, and poorer health.

These computations were implemented in Python using pandas, ensuring that all score variables are numeric and suitable for regression, hypothesis testing, and clustering.

5. Methodology

5.1. Tools

All analysis was performed in Python using Jupyter Notebook. Key libraries included pandas and numpy for data handling, seaborn and matplotlib for visualization, statsmodels and scipy for statistical tests, and scikit-learn for clustering and classification.

5.2 . Analysis pipeline

1. Data cleaning

- Filtered Age.
- Dropped rows with missing values in key variables (gender, scores).
- Encoded categorical variables (Gender, Punctuality, Internet usage).

2. Score construction

- Created Maturity_score, Confusion_score, Distress_score as described in the data section.

3. Exploratory Data Analysis (EDA)

- Plotted histograms for each score.
- Computed correlations between maturity, confusion, and distress.

4. Hypothesis testing

- Simple OLS regressions: Distress ~ Maturity, Distress ~ Confusion.
- Multiple OLS: Distress ~ Maturity + Confusion + Age + Gender.
- Independent samples t-test: distress for males vs females.
- One-way ANOVA: distress across internet usage levels.

5. Clustering

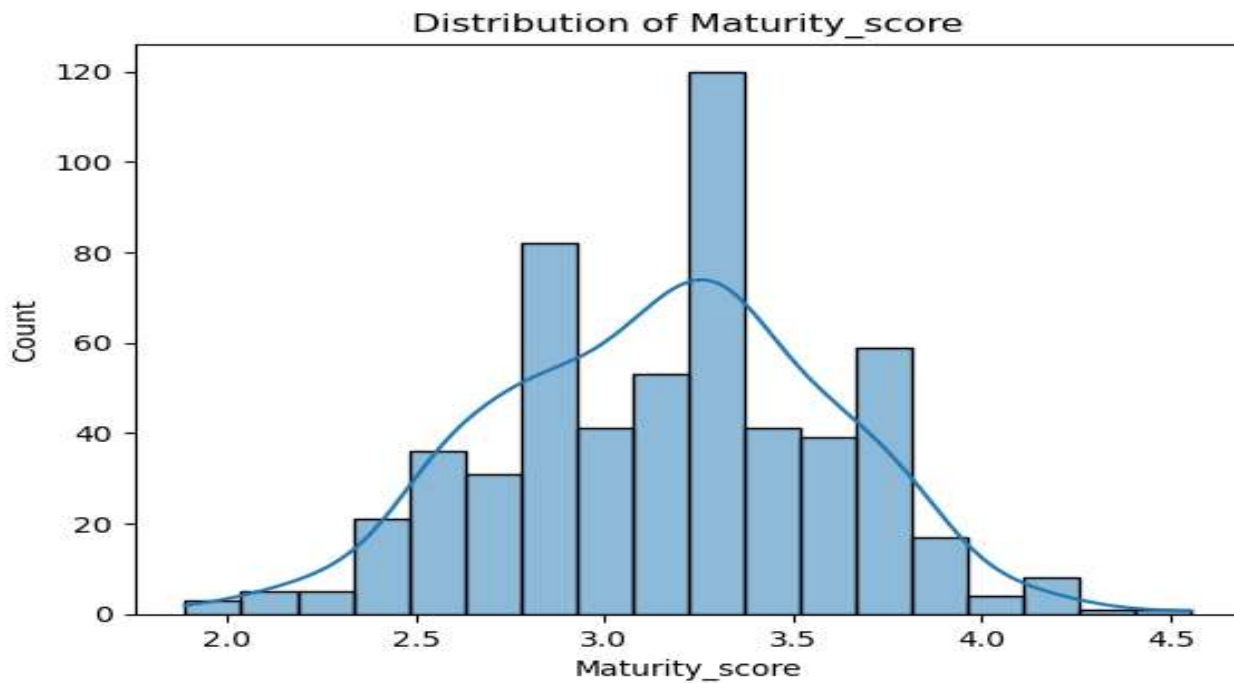
- Applied k-means clustering (k=3) on standardized Maturity_score, Confusion_score, Distress_score to identify profiles.

6. Machine learning

- Logistic regression to classify high vs low distress using scores and basic demographics.

6.1 Exploratory analysis

Fig 6.1



The histogram with the density curve gives a clear picture of how maturity scores are distributed in dataset:

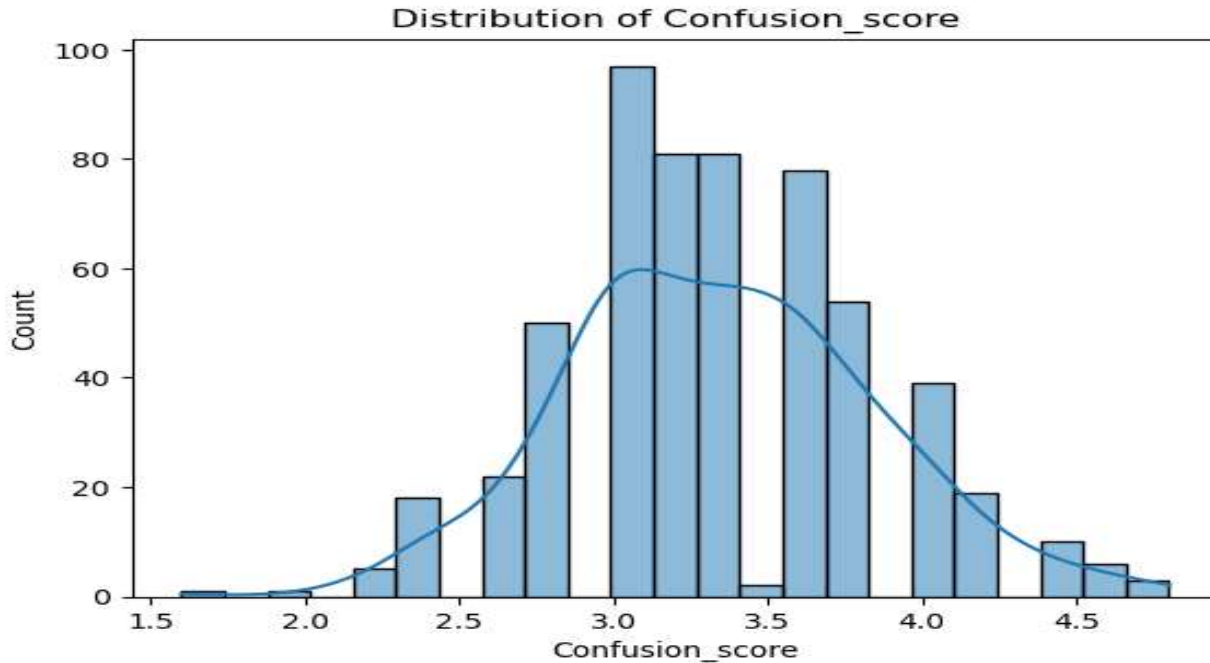
Key Observations

- Range: Scores span roughly from 2.0 to 4.5.
- Center: The peak of the distribution is around 3.0–3.3, which suggests most individuals cluster near this level of maturity.
- Shape: The smooth density curve resembles a normal distribution (bell-shaped), meaning scores are symmetrically spread around the average.
- Highest Frequency: Around a score of 3.3, the count reaches about 120, indicating this is the most common maturity level in your sample.

Interpretation

- The distribution suggests that maturity levels in the group are moderately high, with few people scoring at the extremes (very low or very high).
- Since the curve is close to normal, statistical analyses (like correlations or regressions) that assume normality can be applied.
- The clustering near 3.0–3.3 implies that while there's variation, most individuals are consistently in the mid-to-upper maturity range.

Fig 6.2



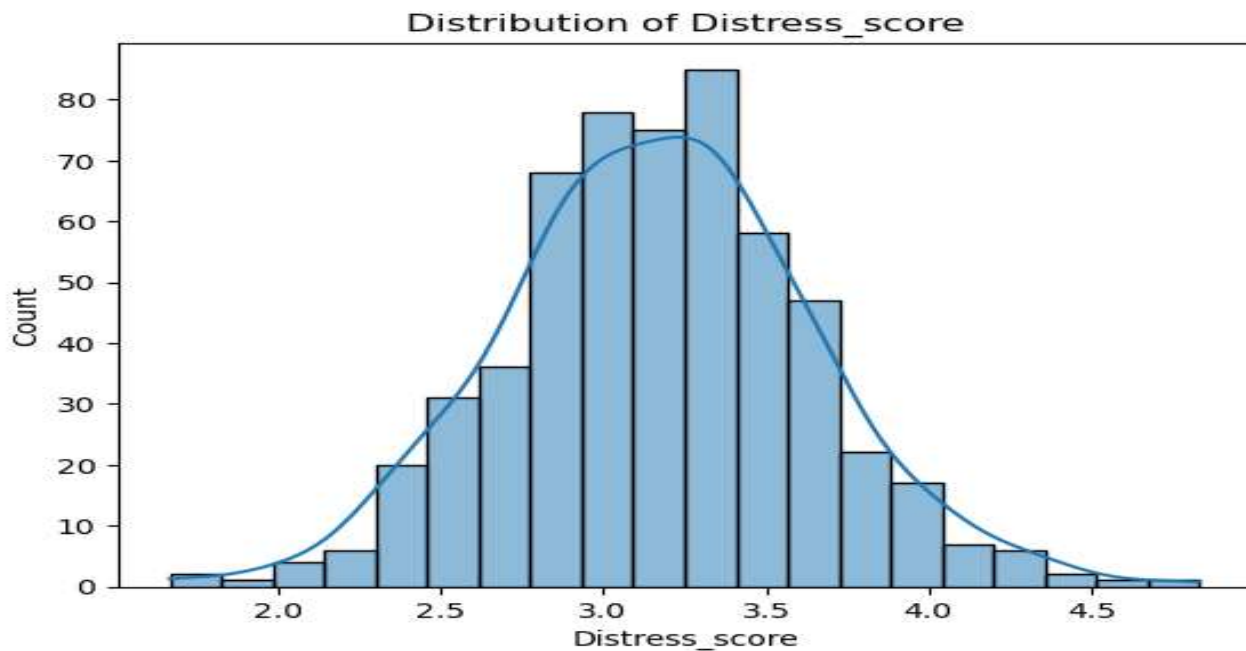
Key Observations

- Range: Scores span roughly from 2.0 to 4.5.
- Center: The distribution is centered around 3.0, with the peak frequency close to 100 counts.
- Shape: The histogram with the density curve shows a bell-shaped pattern, suggesting the scores follow a near-normal distribution.
- Spread: Most individuals cluster around the mid-range (2.8–3.2), with fewer at the extremes.

Interpretation

- The data indicates that confusion levels are moderate for most individuals, with relatively few scoring very low or very high.
- The normal-like distribution suggests that confusion is a common, shared experience across the sample, rather than being skewed toward extremes.
- Compared to the maturity distribution, both seem to center around 3.0, which could imply a balance: as maturity increases, confusion may not disappear but tends to stabilize around a similar midpoint.

Fig 6.3



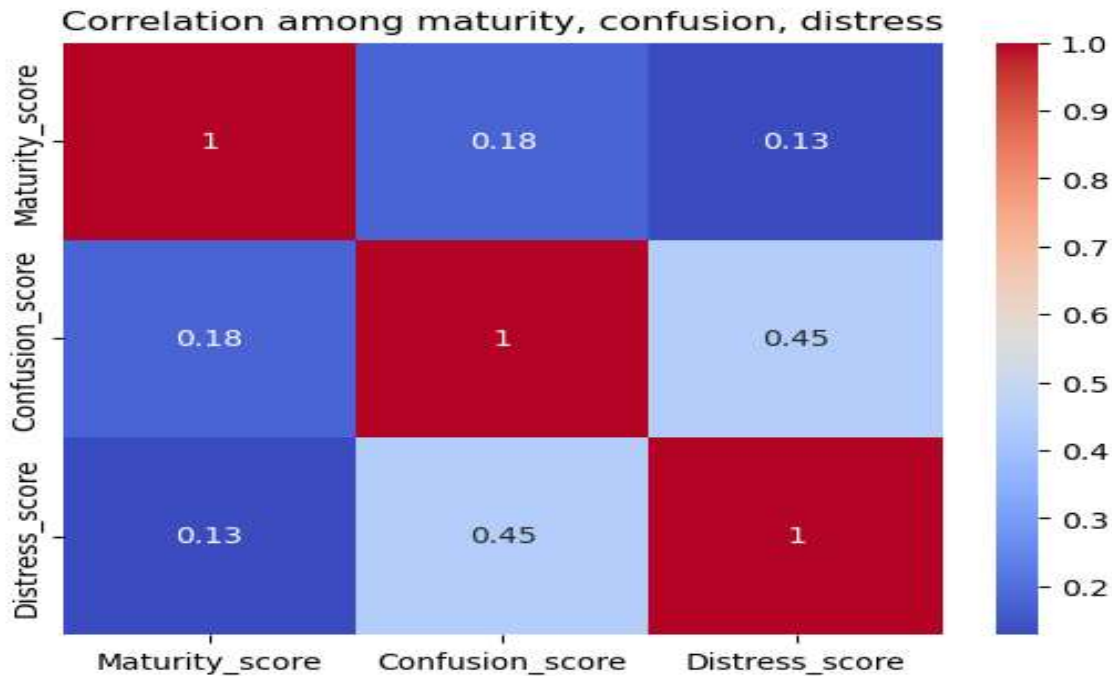
Key Observations

- Range: Scores extend from about 2.0 to 4.5.
- Center: The peak lies around 3.2–3.3, with the highest count.
- Shape: The histogram with the density curve shows a roughly normal distribution, though slightly flatter compared to maturity and confusion.
- Spread: Most individuals cluster in the mid-range, with fewer at the extremes.

Interpretation

- Distress levels in the sample are moderate, with the majority of individuals reporting scores around the mid-point.
- The peak being slightly higher (3.2–3.3) compared to maturity and confusion (centered closer to 3.0) suggests that distress is marginally elevated in this group.
- The normal-like shape indicates distress is a shared experience, not skewed toward only a few individuals.

Fig 6.4



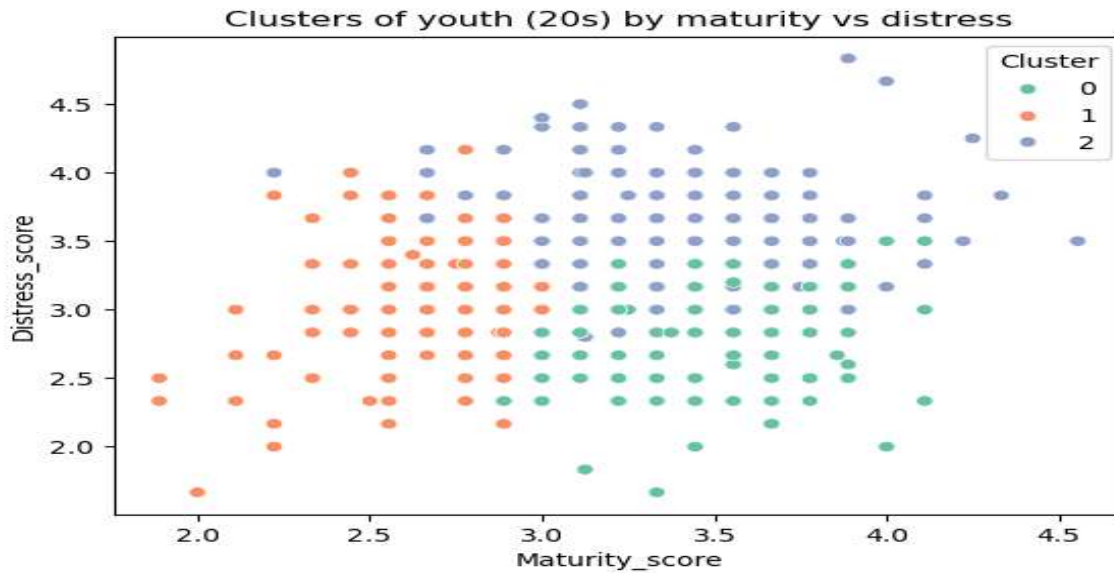
Correlation Values

- Maturity ↔ Confusion: 0.18 → very weak positive correlation.
- Maturity ↔ Distress: 0.13 → very weak positive correlation.
- Confusion ↔ Distress: 0.45 → moderate positive correlation.

Interpretation

- Maturity is largely independent of both confusion and distress. The weak correlations (close to zero) suggest that being more mature does not strongly predict lower or higher confusion/distress levels in this dataset.
- Confusion and distress are linked. The moderate correlation (0.45) indicates that individuals who experience higher confusion also tend to report higher distress. This makes sense psychologically, as uncertainty and difficulty in decision-making often amplify feelings of stress or struggle.
- The heatmap visually confirms this: the cell between confusion and distress is noticeably warmer (closer to red), while maturity's relationships remain cool (blueish).

Fig 6.5



Cluster	Maturity_score	Confusion_score	Distress_score
0	3.4048	3.0678	2.8656
1	2.6709	3.1573	3.0463
2	3.3532	3.7737	3.5629

• **Cluster 0 – High maturity, moderate confusion, lower distress**

- Maturity_score $\approx$ 3.40 (highest among clusters).
- Confusion_score $\approx$ 3.07 (moderate).
- Distress_score $\approx$ 2.87 (lowest among clusters).
- **Interpretation:** Youth in this group tend to be relatively mature and experience moderate confusion, but their distress is relatively low. They may cope better with responsibilities and life pressures, keeping distress under control.

• **Cluster 1 – Low maturity, moderate confusion, moderate distress**

- Maturity_score $\approx$ 2.67 (lowest).
- Confusion_score $\approx$ 3.16 (similar to Cluster 0).
- Distress_score $\approx$ 3.05 (middle).

Interpretation: This group shows the lowest maturity and moderate levels of confusion and distress. They may lack strong planning, reliability, or assertiveness, which keeps their distress slightly above Cluster 0 even though their confusion is not extremely high.

- **Cluster 2 – High maturity, high confusion, high distress**

- Maturity_score ≈ 3.35 (similar to Cluster 0, relatively high).
- Confusion_score ≈ 3.77 (highest).
- Distress_score ≈ 3.56 (highest).
- **Interpretation:** Youth here are relatively mature but highly confused and distressed. They may be responsible and aware, yet face intense uncertainty and life struggles, leading to the highest distress levels. This fits with regression result where confusion has a strong positive effect on distress.

Classification Metrics

Class	Precision	Recall	F1-Score	Support
0	0.56	0.51	0.53	71
1	0.67	0.72	0.69	99
Accuracy			0.63	170
Macro Avg	0.62	0.61	0.61	170
Weighted Avg	0.62	0.63	0.63	170

Confusion Matrix

	Predicted 0	Predicted 1
Actual 0	36	35
Actual 1	28	71

ROC-AUC: 0.664

A logistic regression classifier was trained to predict high distress versus low distress using the constructed survey features. On the test set, the model achieved an accuracy of 0.63 and a ROC-AUC of 0.664, indicating moderate discrimination between the two classes. Class 1 showed better precision, recall, and F1-score than class 0, suggesting the model was somewhat better at identifying higher-distress respondents than lower-distress respondents. The confusion matrix showed 36 true negatives, 35 false positives, 28 false negatives, and 71 true positives, which indicates that the model still makes a substantial number of misclassifications, especially for class 0. Overall, this baseline model demonstrates some predictive value, but further feature engineering or more flexible models may improve performance memory.

6.2 Regression results

1. Multivariate regression of distress on maturity, confusion, age, and gender

OLS Regression Results						
Dep. Variable:	Distress_score	R-squared:	0.223			
Model:	OLS	Adj. R-squared:	0.218			
Method:	Least Squares	F-statistic:	40.17			
Date:	Sat, 18 Jul 2026	Prob (F-statistic):	1.35e-29			
Time:	12:36:06	Log-Likelihood:	-313.31			
No. Observations:	564	AIC:	636.6			
Df Residuals:	559	BIC:	658.3			
Df Model:	4					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	1.4268	0.228	6.271	0.000	0.980	1.874
Maturity_score	0.0631	0.042	1.519	0.129	-0.019	0.145
Confusion_score	0.3597	0.040	9.089	0.000	0.282	0.437
Age	0.0119	0.008	1.520	0.129	-0.003	0.027
Gender_num	0.1464	0.039	3.738	0.000	0.069	0.223
Omnibus:	10.537	Durbin-Watson:	1.934			
Prob(Omnibus):	0.005	Jarque-Bera (JB):	13.889			
Skew:	0.188	Prob(JB):	0.000964			
Kurtosis:	3.670	Cond. No.	290.			

Interpretation

To jointly assess the contributions of psychological maturity, confusion, age, and gender to distress among youth in their 20s, we fitted a multiple linear regression model with distress score as the dependent variable and maturity score, confusion score, age, and gender (0 = male, 1 = female) as predictors.

The model is statistically highly significant overall ($F = 40.17$, $p \approx 1.35 \times 10^{-29}$) and explains about 22.3% of the variance in distress scores ($R^2 = 0.223$, adjusted $R^2 = 0.218$). The estimated regression equation is:

$$\text{Distress_score} = 1.43 + 0.06 \times \text{Maturity_score} + 0.36 \times \text{Confusion_score} + 0.01 \times \text{Age} + 0.15 \times \text{Gender_num}.$$

The coefficient for confusion score is 0.3597 (SE = 0.040, $t = 9.089$, $p < 0.001$), indicating a strong and statistically significant positive association between confusion and distress after controlling for maturity, age, and gender. This means that, on average, a one-unit increase in confusion score is associated with an increase of about 0.36 units in distress score, holding other variables constant, providing clear evidence to reject H_0 .

In contrast, the coefficient for maturity score is 0.0631 ($p = 0.129$), which is not statistically significant at the 5% level in the presence of the other predictors. Therefore, we fail to reject H_0 in the multivariate context, suggesting that once confusion, age, and gender are accounted for, maturity does not have a strong independent linear effect on distress in this sample.

Age has a small positive coefficient (0.0119, $p = 0.129$), and is not statistically significant at $\alpha = 0.05$. However, gender shows a statistically significant effect: the coefficient for `Gender_num` is 0.1464 (SE = 0.039, $t = 3.738$, $p < 0.001$), implying that female youth (`Gender_num` = 1) have, on average, distress scores approximately 0.15 units higher than male youth, controlling for maturity, confusion, and age. This leads us to reject H_0 and conclude that gender is an important factor in distress levels.

Regression of distress on maturity

OLS Regression Results

```
=====
Dep. Variable:          Distress_score    R-squared:                0.017
Model:                  OLS               Adj. R-squared:         0.015
Method:                 Least Squares     F-statistic:           9.573
Date:                  Sat, 18 Jul 2026   Prob (F-statistic):    0.00207
Time:                  12:36:19           Log-Likelihood:       -379.80
No. Observations:      564               AIC:                  763.6
Df Residuals:          562               BIC:                  772.3
Df Model:              1
Covariance Type:       nonrobust
=====
               coef      std err          t      P>|t|      [0.025      0.975]
-----
const          2.7252      0.144      18.881      0.000      2.442      3.009
Maturity_score  0.1402      0.045       3.094      0.002      0.051      0.229
=====
Omnibus:                2.125   Durbin-Watson:           1.926
Prob(Omnibus):          0.346   Jarque-Bera (JB):           2.069
Skew:                   0.056   Prob(JB):                 0.355
Kurtosis:               3.275   Cond. No.                  25.2
=====
```

Interpretation.

To examine the relationship between psychological maturity and distress among youth in their 20s, we fitted a simple linear regression model with distress score as the dependent variable and maturity score as the independent variable.

The estimated regression equation is:

$$\text{Distress_score}^{\wedge}=2.73+0.14\times\text{Maturity_score}.$$

The slope for maturity is 0.1402 with a standard error of 0.045, giving a t-statistic of 3.094 and a p-value of 0.002 (N = 564). Since the p-value is below the 5% significance level, we reject the null hypothesis $H_0H_{0H_0}$ and conclude that maturity score is significantly associated with distress score.

Interestingly, the estimated slope is positive, meaning that in this dataset higher maturity scores are associated with slightly higher distress rather than lower distress. For each one-unit increase in maturity score, distress score increases on average by about 0.14 units, holding other factors constant in this simple model.

Distress on confusion

OLS Regression Results

```
=====
Dep. Variable:          Distress_score      R-squared:                0.199
Model:                  OLS                 Adj. R-squared:           0.197
Method:                 Least Squares       F-statistic:             139.5
Date:                  Fri, 17 Jul 2026     Prob (F-statistic):      6.49e-29
Time:                  11:24:27             Log-Likelihood:          -322.03
No. Observations:      564                 AIC:                    648.1
Df Residuals:          562                 BIC:                    656.7
Df Model:              1
Covariance Type:       nonrobust
=====
               coef      std err          t      P>|t|      [0.025      0.975]
-----
const          1.7433      0.122      14.301      0.000      1.504      1.983
Confusion_score  0.4259      0.036      11.812      0.000      0.355      0.497
=====
Omnibus:          13.133      Durbin-Watson:           1.947
Prob(Omnibus):    0.001      Jarque-Bera (JB):        18.951
Skew:             0.201      Prob(JB):                7.67e-05
Kurtosis:         3.803      Cond. No.                 24.8
=====
```

We fitted a simple linear regression of distress score on confusion score.

The estimated model is

$\text{Distress_score}^{\wedge}=1.7433+0.4259\times\text{Confusion_score}$.

The slope for confusion is 0.4259 (standard error = 0.036), producing $t = 11.812$ and a $p\text{-value} < 0.001$. Because the $p\text{-value}$ is far below the conventional $\alpha = 0.05$ level, we reject the null hypothesis and conclude that confusion score is strongly and positively associated with distress score. Concretely, a one-unit increase in confusion score is associated with about a 0.43-unit increase in distress score on average in this sample. The model explains approximately 19.9% of the variance in distress ($R^2 = 0.199$), indicating that confusion alone is an important predictor but that other variables also contribute to distress.

Hypothesis: Gender Differences in Distress (t-test)

Numerical results

- $t\text{-statistic} = -7.309$

- p-value: 9.83×10^{-13}
- Male mean distress: 3.015
- Female mean distress: 3.299
- **Gender differences in distress.**

We compared mean distress scores between male and female respondents using an independent samples t-test. Under the null hypothesis, mean distress is equal for both groups ($\mu_{\text{male}} = \mu_{\text{female}}$); the alternative hypothesis posits a difference ($\mu_{\text{male}} \neq \mu_{\text{female}}$)

The test yielded a t-statistic of -7.31 with a p-value of approximately 9.8×10^{-13} , indicating an extremely statistically significant difference at any conventional significance level ($\alpha = 0.05$ or 0.01). The average distress score for males was about 3.02, whereas for females it was about 3.30. Since the p-value is far below 0.05, we reject the null hypothesis and conclude that female youth in their 20s report significantly higher distress than male youth in this dataset.

Interpretation

“This result is consistent with many youth mental health studies where female respondents show higher levels of internalizing symptoms (stress, anxiety, depression) than males.”

Hypothesis: Internet Usage and Distress (ANOVA)

Numerical results

- F-statistic: 0.703
- p-value: 0.550
- Group means:
 - Internet_usage_num = 0 (no time at all): mean distress ≈ 3.000
 - 1 (less than an hour a day): ≈ 3.241
 - 2 (few hours a day): ≈ 3.158
 - 3 (most of the day): ≈ 3.150

Internet usage and distress.

We conducted a one-way ANOVA to test whether mean distress scores differ across four levels of Internet usage (no time at all, less than an hour a day, few hours a day, most of the day). The null hypothesis states that mean distress is equal in all groups; the alternative hypothesis is that at least one group mean differs.

The ANOVA produced an F-statistic of 0.70 with a p-value of 0.55. Since this p-value is much larger than $\alpha = 0.05$, we fail to reject the null hypothesis. Statistically, we do not find strong evidence that distress varies by Internet usage level in this sample. The group means range from about 3.00 to 3.24, but these differences are not large enough relative to within-group variability to be statistically significant.

This suggests that, in this dataset, overall distress is not strongly determined by simple self-reported Internet usage category; more detailed measures might be needed to capture any nuanced effects.

Discussion

The present investigation aimed to elucidate the relationships among psychological maturity, confusion, and distress in a sample of young adults aged 20–29, while also examining the roles of gender and internet usage. By constructing composite indices from questionnaire items and applying regression analysis, classical hypothesis testing, and cluster analysis, the study offers a multi-layered picture of youth psychological functioning in this age group.

Confusion as a central correlate of distress

Across both bivariate and multivariate models, confusion emerges as the most salient correlate of distress. In the simple linear regression of distress on confusion, the confusion index is associated with a substantial positive coefficient and accounts for nearly one fifth of the variance in distress scores. This magnitude of explained variance is notably higher than that observed for maturity alone and indicates that confusion is not a marginal factor but a central dimension in understanding distress among young adults.

When maturity, age, and gender are included in a multiple regression model, the coefficient for confusion remains strongly positive and statistically significant, suggesting an independent effect that is not merely an artefact of other predictors. Conceptually, the confusion index aggregates items related to decision difficulty, judgment calls, mood swings, new environments, and life struggles, capturing a broader construct of perceived uncertainty and instability. The empirical findings thus support the notion that confusion—understood as chronic indecision and perceived lack of clarity about life direction—is systematically and meaningfully related to elevated distress in the twenties, a life phase often characterized by transition and heightened role experimentation.

The ambivalent role of psychological maturity

In contrast, psychological maturity exhibits a more ambivalent and limited role in the prediction of distress. The simple regression model indicates a statistically significant association between maturity and distress; however, the effect size is small and the associated R^2 value is very modest, implying that maturity alone explains only a small fraction of the variability in distress. In the multivariate model that includes confusion, age, and gender, the maturity coefficient is no longer statistically significant at conventional levels, indicating that its apparent influence in the bivariate setting may be partly confounded by other variables, particularly confusion.

Maturity in this study is operationalized via items reflecting prioritisation of workload, future-oriented thinking, reliability, keeping promises, punctuality, achievements, and assertiveness. While these characteristics are often assumed to confer resilience, the findings suggest that higher maturity does not straightforwardly translate into lower distress. One plausible interpretation is that greater maturity may entail increased awareness of responsibilities and constraints, which could exacerbate stress when objective circumstances or subjective evaluations remain uncertain. The cluster analysis provides further nuance: the existence of a group with relatively high maturity but simultaneously high confusion and high distress illustrates that maturity may coexist with considerable psychological strain when clarity about life direction is lacking. This underscores the importance of situating maturity within a broader psychosocial context rather than treating it as an inherently protective trait.

Gender differences in distress

The analyses consistently reveal meaningful gender differences in distress. Female respondents exhibit significantly higher mean distress scores than male respondents, as evidenced by both the independent-samples t-test and the regression model controlling for maturity, confusion, and age. The effect is statistically robust and

non-trivial on the scale of the distress index, indicating that gender is an important factor in understanding variations in distress within this age group.

These findings align with a substantial body of literature documenting higher levels of internalising symptoms—such as anxiety, sadness, and perceived stress—among women and girls relative to their male counterparts. Although the present study is not designed to identify causal mechanisms, the pattern suggests that gender-related socialisation processes, differential exposure to stressors, or variations in coping strategies may contribute to the observed disparities. For educational institutions and early-career workplaces, this evidence highlights the need to consider gender-sensitive approaches in designing mental health outreach, support services, and prevention programmes.

Internet usage and distress: null effects in coarse measures

By contrast, the analysis finds no statistically significant differences in distress across the broad categories of internet usage examined. While mean distress scores vary slightly between usage levels, the one-way ANOVA does not indicate significant group differences at the conventional 5% level. On the basis of these results, simple self-reported daily usage categories appear to be relatively weak predictors of distress compared to psychological indices such as confusion and maturity.

However, this null finding should be interpreted with caution. The operationalisation of internet usage is coarse, capturing only approximate duration rather than the nature, context, or content of online activity. It is plausible that the qualitative aspects of internet use—such as engagement with social media versus educational content, exposure to negative or supportive interactions, and timing of use—are more relevant to distress than the total time spent online. Moreover, the influence of internet usage may depend on individual differences and environmental factors not explicitly modelled here. The present study therefore suggests that, within the limits of the available measures, internet usage is less informative than confusion and gender as a predictor of distress, but it does not rule out more nuanced relationships.

Psychological profiles derived from cluster analysis

The cluster analysis complements the regression and hypothesis-testing results by identifying distinct psychological profiles based on the joint distribution of maturity, confusion, and distress. One cluster is characterised by relatively high maturity, moderate confusion, and comparatively low distress, and may be interpreted as a group with relatively effective coping and adaptation. A second cluster displays lower maturity, moderate confusion, and intermediate distress, suggesting that limited maturity may be associated with somewhat elevated distress even in the absence of extremely high confusion.

The third cluster, combining relatively high maturity with the highest confusion and highest distress, is particularly informative. It illustrates that maturity and distress are not inversely related in a simple linear fashion. Rather, mature individuals may experience substantial distress when faced with persistent uncertainty and life struggles, possibly because their awareness of responsibilities and expectations amplifies the psychological impact of unresolved confusion. Together, these clusters support the regression-based conclusion that confusion constitutes a central axis of variation in distress and demonstrate how different configurations of maturity and confusion correspond to distinct distress profiles.

Implications for research, practice, and analytics

From a research perspective, the study illustrates the utility of constructing theoretically motivated indices from survey data and integrating multiple analytic approaches—exploratory analysis, regression, classical tests, and clustering—within a coherent quantitative framework. The results also underscore the importance of attending

to psychological constructs that capture uncertainty and decision difficulties, in addition to more traditional measures such as demographic characteristics or behavioural indicators.

From a practical standpoint, particularly in educational and early-career settings, the findings suggest that interventions aimed at reducing distress should consider confusion as a primary target. Programmes focused on career guidance, structured decision-making support, mentoring, and psychoeducation about coping with uncertainty may be particularly relevant for youth in their twenties. Given the observed gender differences, such interventions may need to be designed and delivered with explicit attention to gender-specific experiences and needs.

In the context of analytics roles, the project demonstrates an end-to-end workflow that is directly applicable to data-driven decision-making: feature engineering from complex survey data, application of appropriate statistical models, validation of hypotheses, segmentation of the population through clustering, and translation of quantitative outputs into interpretable insights. This illustrates how statistical and machine learning tools can be deployed not only to produce numerical results but also to inform substantive understanding and practical recommendations.

Limitations and directions for future research

Several limitations should be acknowledged. First, the data are cross-sectional and based on self-report, which restricts the ability to draw causal inferences and introduces potential measurement biases. Second, the composite indices of maturity, confusion, and distress—although conceptually grounded—are not derived from standardised psychometric instruments, which may limit their comparability with other studies and the precision with which they capture underlying constructs. Third, the sample is drawn from a non-Indian context, and cultural differences may influence both the expression and the reporting of maturity, confusion, and distress.

Future work could build on the present study by collecting longitudinal data to examine trajectories of maturity, confusion, and distress over time, particularly around key transition points such as examinations, graduation, and entry into the labour market. Employing validated psychological scales would strengthen construct validity, and focusing on specific populations—for example, university students in India—would enhance contextual relevance. More detailed measurement of digital behaviour, including types of online activity and social media use, could clarify the role of internet engagement in psychological outcomes. Finally, the use of more flexible modelling techniques that can capture non-linear effects and interactions, while retaining interpretability, would permit a richer understanding of how maturity, confusion, gender, and contextual factors jointly shape distress in early adulthood.

In sum, the study provides convergent evidence that confusion is a central quantitative and conceptual correlate of distress among young adults, that gender differences in distress are substantial, and that the role of maturity is contingent on the broader psychological and situational context. These conclusions contribute to the empirical literature on youth mental health and simultaneously illustrate the application of statistical and machine learning methods in an academically rigorous manner.

